IB Subject: Mathematics

Extended Essay

Topic: Evaluating South Korean pollsters with the 538 statistical methodology

Research Question: How, and to what extent can the 538 Pollster rating system be modified on South Korean Pollsters for extensive ratings and rankings?

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Introduction and rationale

The FiveThirtyEight (FiveThirtyEight), also known as 538, is a website renowned for data-driven analysis on political polling within the USA. Founded by Nate Silver, they gained prominence in 2008 as they predicted the winners of the presidential election in 49 of the 50 states. 538 is also respected for rating pollsters, combining various empirical and impartial evidence to create accurate scores that can forecast the 'goodness' of a pollster with a rating out of 3.

I came across 538 listening to a PBS podcast about American elections. The existence of such an organization made me wonder about Korean pollsters, and how they may be ranked, but surprisingly, I was unable to find a Korean counterpart of 538. Attracted by the statistical challenges and real world implications of mathematics waiting to be explored, I decided to attempt to create a pollster ranking system based on that of 538 for my Extended Essay.

This extended essay will attempt to answer the question, "How, and to what extent can the 538 Pollster rating system be modified on South Korean Pollsters for extensive ratings and rankings?", attempting to replicate their methodology on pollsters in South Korea. Political polling is a crucial part of democratic politics and election processes, as it provides insight into political opinion that influences political strategies, and even the decisions of elected government officials. Albeit the differences inherent within Korean and US political, voting and polling systems and the difficulties they pose in working with the initial 538 system, the research will hopefully provide a detailed and evidence-based understanding of what makes a good or bad pollster that can help voters, parties and candidates make informed decisions within Korea.

Section I: Initial Data Collection

The 538 Methodology

The 538 combines figures about not just empirical accuracy, but also bias, transparency and methodological accuracy to rate pollsters (Morris). Historical accuracy figures from the past are not enough to rate pollsters as they are inherently outdated information that shows how pollsters have done, not a predictive measure of pollsters' future performance. There is therefore a need for other measures that reflect the pollster's reliability and consistency which reflects its trustworthiness in the future. Hence, transparency, bias and methodological reliability are needed to reflect these qualities in pollsters.

Constructing the Database

Source

In creating the database of polls from multitudes of pollsters for this research, the archive of the National Election Commission (중앙선거관리위원회) was used to collect data about past elections in South Korea. Following the data collection system of the 538 methodology, only polls that met the following conditions were considered:

1. Conducted after 1998
2. Surveying Presidential, General or Senate or House elections
3. Having a median field date within 31 days (14 only for presidential primary elections)

All other data collection policies listed by the 538 were followed (Morris). Because of later calculation processes, polls were collected from chosen specific elections, instead of being

picked individually. So for the purposes and scope of this extended essay, just 71 polls that surveyed the 19th and 20th presidential elections, conducted respectively in 2017 and 2022, assembled the database to be used for calculations.

Collected Data

The assembled polls were from 10 separate polling organizations, which are referred to as pollsters: Jowon C&I (조원씨앤아이). Kantar Korea (칸타코리아), Realmeter (리얼미터), Hankook Research (한국리서치), Research View (리서치뷰), Ace Research (에이스리서치), Gallup Korea (한국갤럽조사연구소), Research & Research (리서치앤리서치), PNR, and KSOI. The following data was collected about each pollster and their polls:

Median field date

The median field date, defined as the median date in which the poll was conducted over, and how far it is from the actual election date is an important figure in calculating the 'expected error' for each poll. Polls conducted longer before the election are more subject to variability and potential inaccuracies due to changes in public opinion, news events and such dynamic factors, and are therefore expected to be less accurate than polls conducted shortly before election dates.

Error and bias

Raw error was measured and recorded as the difference between the predicted and actual figure for the **turnout difference** between two candidates with the most votes. Raw bias is simply raw error, but positive or negative depending on whether the poll overestimated republicans or democrats. Since there are more than two parties in Korean politics, 'Republicans'

and 'Democrats' will be replaced by the relatively conservative/right wing and liberal/left wing parties, respectively, within this research. These values are used to calculate figures for both the accuracy and precision of pollsters.

Affiliation to a trustworthy pollster association

Each pollster's affiliation to the KORA, a Korean counterpart to the AAPOR Transparency initiative and the Roper center the 538 considers as a boost to pollsters' transparency and methodological reliability, was recorded. The AAPOR and Roper center are well respected US organizations that promote transparency and ethical standards in public opinion polling. The 538 found pollsters that are affiliated with either generally showed lower error rates than those who weren't. Researching the KORA, their size, credibility, and terms, it was decided to be an effective replacement for the AAPOR transparency initiative and Roper center in that it receives additional private information from member pollsters and monitors them, and continuously attempts to propagate scientific and accurate research.

Transparency criterias

Ten criteria for transparency, listed by the 538, were used to measure a large part of transparency. The number of the criterias each poll successfully fulfills is the main measure of transparency for the conducting pollster.

Polling Methodology

All polls used ARS (Automated Response System, a method of polling done with automatic interview machines), phone interviews or a combination of both in their polling. Since recent studies from the National Election Committee (Koo) and statements from the KORA (Jeong, Lee) has evaluated that ARS is not a scientific method of polling due to low response rates and a possible volunteer bias, pollsters were penalized in the proportion of ARS polling of their every poll.

Sample size

The sample size (the total number of participants within a poll) was also recorded for every poll. This figure is crucial in calculation processes, in determining the weight of each poll and calculating the expected error for each poll.

Poll number	3700	3702	3703	3704	3708	3709	3710	3712	3714	
Pollster	리얼미터	리얼미터	한국리서치	한국리서치	리서치뷰	한국갤럽조사연구	리서치앤리서치	리얼미터	에이스리서치	리
Median Field Date I	14	13.5	14	13	13	13	13	11	10	
Difference Predictic	31.4	31.6	28.8	23.6	30.2	28	29.9	25.9	20.3	
Raw Error	14.3	14.5	11.7	6.5	13.1	10.9	12.8	8.8	3.2	
Raw Bias	14.3	14.5	11.7	6.5	13.1	10.9	12.8	8.8	3.2	
Transparency 1	1	1	1	1	1	1	1	1	1	
Transparency 2	1	1	1	1	1	1	1	1	1	
Transparency 3	1	1	1	1	1	1	1	1	1	
Transparency 4	1	1	1	1	1	1	1	1	1	
Transparency 5	1	1	1	1	1	1	1	1	1	
Transparency 6	1	1	1	1	1	1	1	1	1	
Transparency 7	1	1	1	1	0.5	1	1	1	1	
Transparency 8	1	1	1	1	1	1	1	1	1	
Transparency 9	0	0	0	0	0	1	0	0	0	
Transparency 10	0	0	0	0	0	0	0	0	0	
Methodology	80% ARS, 20% I	80% ARS, 20% I	Phone Interview	Phone Interview	ARS	Phone Interview	CATI	80% ARS, 20%F	ARS	5
KORA Affiliation	0	0	3	3	0	3	0	0	0	
Sample size	1520	1015	1000	1000	1418	1006	1007	1523	1023	
Confidence Interval	95% +-2.5%	95% +-3.1%	95% +-3.1%	95% +-3.1%	95% +-2.6%	95% +-3.1%	95% +-3.1%	95% +-2.5%	95% +-3.1%	9

Figure 1, excerpt of database created by the author

Section II: Calculations

Using the collected database, the measures for each pollster's empirical accuracy and precision, and also their transparency and methodological reliability is calculated as figures named POLLSCORE and Transparency Score. These figures are then combined to produce singular ratings on a scale of zero to three, that reflects the overall quality of the pollster.

Accuracy and Precision - POLLSCORE

POLLSCORE (Predictive Optimization of Latent skill Level in Surveys, Considering Overall Record, Empirically) is calculated through multiple steps, modifying raw error and bias to reflect various aspects of empirical accuracy.

Raw error & Bias

Raw error and bias is calculated like shown in Section I. These figures serve as the foundations for measuring accuracy and precision, but are misrepresentative or biased when interpreted alone. Therefore, adjustments must be made to this figure.

Expected Error - Voter volatility and statistical uncertainty

One key aspect to be considered is the statistical uncertainty and voter volatility due to pure chances within the polling process. 538 mentions that it accounts for this uncertainty by calculating 'expected error' an inherent error a poll is expected to have for these reasons, by running a multilevel regression model combining four variables:

- the implied standard deviation of a poll from its sample size,
- the square root of the number of days between the median field date of the poll and the election,
- plus variables for the cycle the poll was conducted,
- and finally the type of election it sampled (Morris).

The fourth variable is irrelevant in the scope of this investigation, since the aim is to create a ranking between pollsters, and the entire database only consisted of polls about presidential elections. The third variable was deemed to be too extensive within the scope of the investigations.

With this, a hypothetical equation for expected error considering the two other variables is created:

$$expected\ error\ =\ a\sqrt{date}\ +\ b\sigma$$

The *date* is simply data that was already collected for every poll. Using formulas already learnt, an attempt was made to derive standard deviation (σ) from the confidence intervals of each poll, since this information was available for all Korean polls.

$$CI\ =\widehat{p}\pm\ z\sqrt{\frac{\widehat{p}\ (1-\widehat{p})}{n}}$$

Confidence intervals are parameters in which the population means are expected to be contained under a certain probability called confidence, calculated from the formula above for *CI*, where

- $\hat{p}$ represents the sample proportion,
- n is the sample size,
- and z stands for the critical value, determined by the confidence level (Watts).

FiveThirtyEight's methodology references the "implied standard deviation of a poll from its sample size." Statistically, this refers to the standard error (SE) of the sample proportion which quantifies how much $\hat{p}$ varies across repeated samples.

SE can be automatically calculated from the details of a confidence interval due to its relationship with the margin of error (ME) - the width of the confidence interval - and the critical value. SE can be derived from the confidence interval using the formula:

$$\sigma_{\hat{p}} = SE = \sqrt{\frac{\hat{p}\,(1-\hat{p})}{n}}$$

$$SE = \frac{ME}{2z}$$

Therefore, this formula could be used to calculate the expected error of polls in the database once the coefficients for each variable, a, and b had been calculated.

An attempt was made to use the 538 database of polls and pollsters, specifically, each poll's expected error, median field date and standard deviation. With this data, the Desmos online

calculator can be used to create sliders for *a* and *b*, and fit them to a sample of these figures from the database. However, while the median field date and expected error could be found, standard deviation, nor confidence intervals or any figure necessary to calculate it was unavailable in the database.

However, considering the relationship that can be seen from the formula,

$$\sigma \propto \frac{1}{\sqrt{n}}$$

, and also the 538's emphasis in the wording for this variable in how they state standard deviation is 'implied by its sample size', a decision was made to use $\frac{1}{\sqrt{n}}$ in the place for standard deviation to represent the relationship between sample size and standard error. This would change the previously established hypothetical equation to look like this

$$expected\ error = a\sqrt{date} + b\frac{1}{\sqrt{n}},$$

and result in a different value of *b* than if standard deviation was used.

Since figures for standard error were only available at pollster levels, three pollsters with manageable numbers of polls (ranging from 2 to 4), Knowledge Networks, POS/Hamilton Strategies, and Mckeon&Associates, were selected as samples out of the 538 database of pollsters (pollster-stats-full) (Fuong), and collected data about polls from these using the 538 database of raw polls (raw_polls) (Fuong).

Because polls are weighted by how old they are, the number of years since 2023 (database date) was recorded for each poll, in addition to their sample size and median field date figures.

	Year	Year before calculations (*d*)	Median Field Date (*x*)	Sample Size (*y*)	Meets condition for collection	Weight (*w*)
1	2008	15	55	1546	x	0.1041062417
2	2008	15	27	1528		0.1041062417
3	2008	15	13	1537		0.1041062417
4	2003	20	21	702		0.04897438882
5	2003	20	7	702		0.04897438882
6	2004	19	53	959	x	0.05694696374
7	2004	19	18	968		0.05694696374
8	2006	17	28	1058		0.0769969764

Figure 2, sample of collected data from 538 about Knowledge Networks. Spreadsheet created by author

Notably, however, of the three pollsters analyzed, Knowledge Networks had eight polls instead of four like the 538 database stated. While all six of the eight qualifying polls were used for later calculations, this raised concern in the accuracy of later results, as it implied 538's calculations for their expected error for this pollster may be coming from a separate database or a different selection system, and potentially invalidates my regression.

Taking note of the 538's wording that polls' weight decay by 'about 14% each year', weight was given to each poll by the 86% to the power of years before calculations by 538. Hence, using this data, I used heuristic tuning using sliders on Desmos to calculate the coefficients *a* and *b*.

Firstly, two lists about the median field date and sample size were created, *x* and *y*. Then, using the formula mentioned above, a list of expected error for *each poll*, *e* was created with undefined variables *a* and *b*:

$$e = ax + b\frac{1}{\sqrt{y}}$$

Next, these figures are combined using the weighting system mentioned above. On the table below, the lists *d* and *w* represent the date, and the weighting each poll therefore receives, respectively. Combining the expected error, *e* and weighting *w*, final *weighted expected errors for each poll*, *f* can be calculated:

$$f = ew$$

x_1	y_1	e_1	d_1	w_1	f_1
27	1528	4.482598	15	0.10410624	0.46666643
13	1537	4.1543816	15	0.10410624	0.43249705
21	702	5.9966648	20	0.048974389	0.29368299
7	702	5.6092999	20	0.048974389	0.27471203
18	968	5.174736	19	0.056946964	0.2946855
28	1058	5.1964124	17	0.076996976	0.40010804

Figure 3, sample of regression calculations, for Knowledge Networks

(Created by author using Desmos)

Then, the expected error for the pollster can be found. For the expected error of Knowledge networks, *k* for example:

$$k = \frac{\Sigma f_1}{\Sigma w_1}$$

The same is done for the other two pollsters. These figures for expected error for pollsters can then be compared to the actual data the 538 published. Coefficients *a* and *b* can finally be found by finding a combination of the two that brings *k*, *p*, and *m* closest to actual.

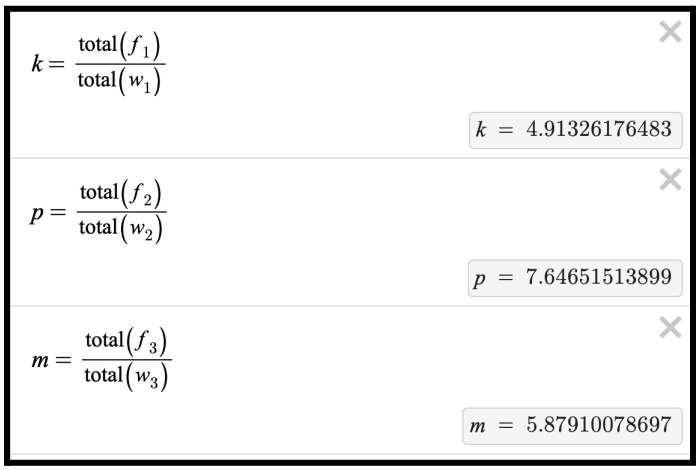


Figure 4, equations for k, p*, and* m

To measure the 'closeness' of calculated expected error to actual, the formula for error rate, learned in the IB curriculum, was used:

$$\delta = \left| \frac{v_e - v_a}{v_e} \right|$$

Best-fit regression functions in the desmos calculator could have been used to find *a* and *b*, the coefficients for date and inverse sample size for *e* such that the combined error rate would be the smallest. However, since the data set being worked with was small and and somewhat unreliable due to reasons explained, a decision was made against this methodology since it may

lead to overfitting and misleading results. Instead, the author's intuition and trial and error was to be used to find accurate values for *a* and *b*.

After setting a function for the combined value of the absolute error rates for each of the 3 pollsters' error rates, the slider function in Desmos was used to manipulate figures for *a* and *b*, and find a combination of the two that produced a small combined error rate and also be reasonable. The range of *a* and *b* were set from 0.15 to 0.50, and 100 to 200 respectively.

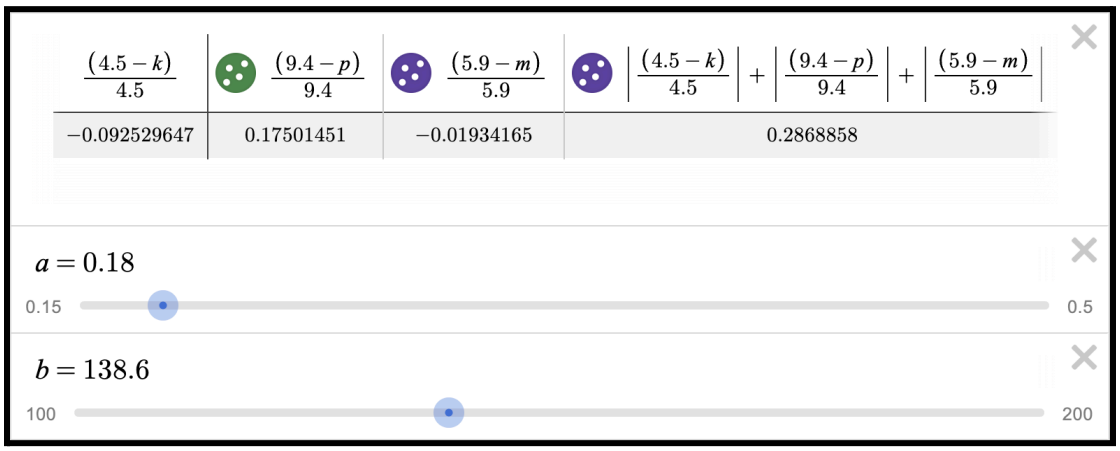


Figure 5, example of trials with higher error #1

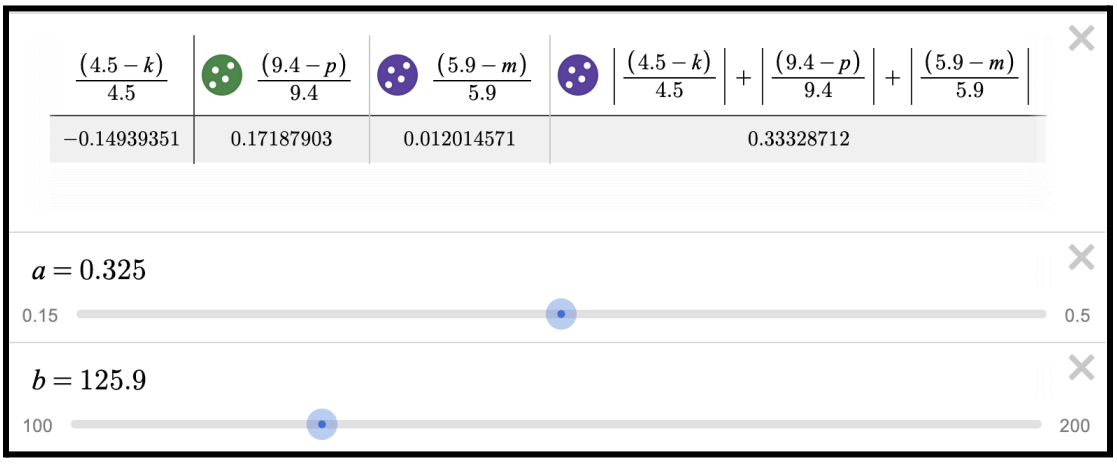


Figure 6, example of trials with higher error #2

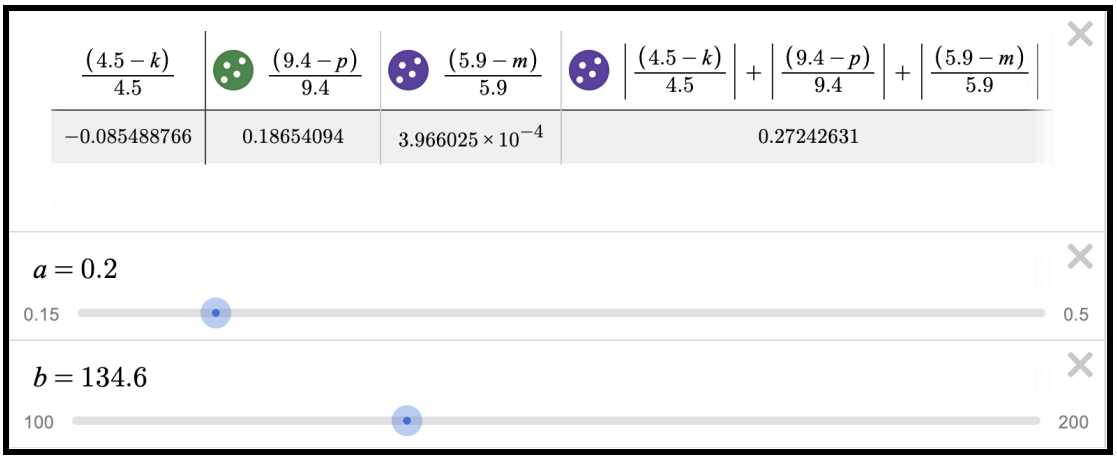


Figure 7, final values for a *and* b

Resultantly, values for *a* and *b* were determined to be 0.2000 and 134.6, respectively. Hence, the formula for expected error is redefined as:

$$expected\ error\ =\ 0.2\sqrt{date}\ +\ 134.6\frac{1}{\sqrt{n}}$$

and expected error for each poll can be calculated.

Excess error & Bias

With expected error, excess error, which is error unaccounted by expected statistical volatility, can be calculated by subtracting expected error from raw error. Excess bias is simply raw error, since pollsters are not expected to be biased at all.

Adjusted Error

One last factor to be considered is race difficulty. Some elections are, by reasons unrepresented by numerical statistics, harder to predict and/or more volatile than others. Since pollsters poll multiple elections and in different repetitions, it is necessary to account for race difficulty in order to not be biased against pollsters that polled many difficult elections and vice versa. The same goes for bias.

The *average excess error and bias* of each election must be referenced in order to do this. Weighting the adjusted error and bias for every poll in each election by the square root of their sample size for reasons mentioned above and averaging these figures, the average excess error and bias of each election is calculated.

	19th Presidential Election	20th Presidential Election
Average excess error	3.49	–1.20
Average excess bias	7.80	-2.05

Once this is calculated, the *adjusted error and bias* is calculated for each poll by calculating how much its excess exceeds the average error and bias of the election it is surveying, subtracting average from excess. However in hindsight I realize a possible limitation to this approach in that it punishes polls with bias figures lower than expected, which is unfair and counteractive for polls that managed to be relatively unbiased and impartial. This problem is especially relevant within my small dataset.

Predictive Error

Predictive error is a final step to "turn a pollster's Adjusted Error and Adjusted Bias into a prediction" (Morris). This process involves multiple unspecified "regressions models" and accounting for 'herding', where pollsters modify their results to match other polls. The process was determined to be too extensive and ambiguous for the scope of the research, so this process is excluded and adjusted error and bias is used in the place of predictive error/bias.

POLLSCORE

Finally, using the calculated adjusted error and bias figures for every poll, the *pollster adjusted error/bias* is calculated as an average of its performance on its polls. Each poll is weighted by a weight decaying by 14% each year, as explained. Adjusted bias on the pollster level is converted to an absolute value, because it matters not to which direction a pollster is biased towards, but the degree it is.

POLLSCORE is then calculated for each pollster by averaging its adjusted error and bias figures together.

Transparency and reliability

The transparency of pollsters are derived relatively simply, as the weighted(by time) average of their polls' mentioned qualifications for 10 criterias. This accounts for 70% of the transparency score. Also, the pollster's affiliation to KORA accounts for 20%, as an additional proof of transparency and scientific methodology. The 538 contrastingly gives a weight of 30% for AAPOR/Roper affiliation, but the trustworthiness of the institute is not as clear for the researcher as it was for 538, the author found it for, and within, the investigation itself.

However, it was noted after the initial collection of this data that every poll within the database had criterions 1-8 filled out perfectly, or very close to that. This may have been due to regulations and policies the archive that was used for data collection, the National Election Committee, enforced on pollsters to be published on their website. This somewhat weakened transparency as a discriminatory measure, since a large part of the score was effectively equivalent across all polls and pollsters. In finding a suitable measure to counteract this, and award pollsters with accurate and scientific methodologies, each poll's *polling methodology* was noted. 10% of transparency scores were based on how much the pollster avoided using ARS and instead used phone interviews for their polling, as a proportion.

So the final transparency score can be calculated as:

$$Transparency = 0.7(criteria\ score) + 2(affiliation) + 1(phone\ interviews)$$

Where *criteria score* is a raw score out of 10, *affiliation* is binary, and *phone interviews* is expressed as a percentage, giving a score out of 10.

Organized Results

As a result of these calculations, POLLSCORE and transparency figures can be found for each pollster:

Pollster name	POLLSCORE	Transparency
Jowon C&I	0.81	5.85
Kantar Korea	4.26	8.6
Realmeter	1.31	6.17

Hankook Research	1.87	8.81
Research View	2.15	5.71
Gallup Korea	-0.59	9.18
Research & Research	-0.14	7.71
PNR	2.44	5.6
Ace Research	0.61	5.6
KSOI	0.15	6.95

Table 1, POLLSCORE and Transparency figures for Korean Pollsters
Figures simplified to 2 decimal places for visibility

A lower POLLSCORE value indicates better performance, with a perfect pollster expected to have a POLLSCORE of 0, meaning it is entirely free of error and bias. However, due to the way adjusted error and bias are calculated, a pollster can achieve a negative POLLSCORE if it performs exceptionally well compared to other pollsters in the elections it surveyed.

In contrast, Transparency Scores range from 0 to 10, where higher values indicate greater transparency and methodological reliability.

Section III: Final Ratings

Calculating a final rating

For the final rating to put all of POLLSCORE and Transparency values together, 3 possible ways were considered. This was the final step after other calculations were done, and the objective was not just to completely copy the 538 methodology and produce results for pollsters exactly how it would be as a part of their database, but to create a comprehensible ranking system with differentiated pollsters. Therefore, while it could have been done, the database of

538 pollster' POLLSCOREs and weighted average Transparency scores was not studied and used to find coefficients for a hypothetical equation, like done for expected error with DESMOS. Instead 3 different approaches were used to create 3 sets of pollster ratings and ranks. Each were a version of this equation:

$$Rating = a(\frac{10-POLLSCORE}{10}) + b(\frac{Transparency}{10})$$

The sum of *a* and *b* is 3; both components of the equation are supposed to max out at 1 and have a minimum of 0. However while transparency's limits are clearly bound, the POLLSCORE could go above 10 or below 0, which makes the component escape the stated boundaries and create a rating lower or higher than 0 or 3. The latter did happen in the research with the pollsters Gallup Korea and Research & Research with POLLSCORES -0.59 and -0.14, and is a possible limitation to the research.

Approach 1 - POLLSCORE heavy

One approach was to simply set *a* and *b* myself, putting more emphasis on POLLSCORE by making *a* = 1.8 and *b* = 1.2. If one values historical accuracy above all else, this method is reasonable.

Approach 2 - Transparency heavy

Another approach, much like approach 1, was to set *b* 1.8 and *a* 1.2 This method places more significance on transparency and precision, and may be reasonable for those who do not view past performance as very predictive of future performance.

Approach 3 - Varying coefficients

One other approach, considered due to the small size of the database, was to weight POLLSCORE differently depending on how much of their polls were in the database. Since empirical accuracy is less controllable and predictable than transparency and methodological accuracy, higher numbers of sample polls would reduce the uncertainty of luck in POLLSCORE, and vice versa.

$$a = 2.4 - \frac{1.2}{e^{0.5(n-1)}} \qquad b = 3 - a$$

This equation sets the *a*'s boundaries from 1.2 to 2.4. If the pollster had just 1 poll,

$$a = 2.4 - \frac{1.2}{e^{0.5*(1-1)}}$$

$$a = 2.4 - \frac{1.2}{1}$$

$$a = 1.2$$

and if the pollster had an infinite amount of polls,

$$a = 2.4 - \lim_{x \to \infty} \frac{1.2}{e^{0.5(z-1)}}$$

$$a = 2.4 - \lim_{x \to \infty} \frac{1.2}{x}$$

$$a = 2.4 - 0 = 2.4$$

Results

Using these different rating methodologies, the final ratings for the pollsters are organized down below.

Pollster name	Rating 1 (Accuracy heavy)	Rating 2 (Transparency heavy)	Rating 3 (Varying weights)	Average rating
Gallup Korea	3.01	2.92	3.05	2.99
Research & Research	2.75	2.6	2.83	2.73
KSOI	2.61	2.43	2.7	2.58
Hankook Research	2.52	2.56	2.49	2.52
Realmeter	2.3	2.15	2.43	2.29
Jowon C&I	2.36	2.16	2.39	2.30
Ace Research	2.36	2.13	2.25	2.25
Research View	2.1	1.97	2.09	2.05
Kantar Korea	2.06	2.24	2.03	2.11
PNR	2.03	1.91	1.98	1.97

Table 2, ratings for Korean pollsters

Figures simplified to 2 decimal places for visibility

Gallup Korea seems to be the best pollster within the scopes of this research, scoring the best on all three rating systems. PNR consistently holds the worst ratings, with minimal variation between rating systems. There generally seems to be more variability around the middle range, with pollsters that have average ratings above 2.1 and below 2.8.

Testing the model

The model was tested against the actual results of a recent 22nd General Election, which took place in 2024 May, Korea. Data was collected about polls that surveyed public support for the proportional representation for congressmen. The goal was to create a rank for pollsters on their accuracy in this election and evaluate the predictive rankings using the spearman's rank correlation coefficient.

Polls ranging from 7 to 13 days before the election date were picked. There were only 7 pollsters that met these criteria, with others going out of service or not participating.

Of these pollsters, accuracy was measured mainly like raw error - one difference was the decision to include the third-place party as well, since they received a significant turnout (24.3%), since a 'good' pollster is expected to represent public opinion fully including different dynamics. Since there was only 1 poll from each pollster analyzed, the calculation was relatively simple with no need for steps like in deriving POLLSCORE. The actual results were:

People Power Party (국민의미래)	Democratic Alliance of Korea (더불어민주연합)	Rebuilding Korea Party (조국혁신당)
36.7	26.7	24.3

and the calculated rankings in accuracy, of the 7 relevant pollsters is calculated and organized into a pollster/ranking format is:

	Ranking by rating 1	Ranking by rating 2	Ranking by rating 3	Ranking by 22nd General Election
Gallup Korea	1	1	1	1
Research & Research	2	2	2	2
Hankook Research	3	3	3	3
Ace Research	4	6	6	4
Jowon C&I	5	4	5	6
Realmeter	6	5	4	7
Research View	7	7	7	5
Spearman's rank correlation coefficient with actual	0.893…	0.714…	0.679…	

Table 3, pollster ranking data by each systems and the 22nd General Election
Figures not rounded, but cut off at the third decimal place

Using this reorganized data, the author's GDC is used to calculate the spearman's rank correlation coefficient between each rating and the results in the 22nd general election, shown in the bottom of the table above. The spearman's rank correlation and coefficient measures the degree of the correlation between two ranked variables, in this case the ranks of pollsters' ratings and their ranks in accuracy in the mentioned election. Therefore, this coefficient, R_s, gives insight into the accuracy of the three ranking systems for pollsters created.

Conclusion

This Extended Essay successfully developed and tested a mathematical model for evaluating South Korean pollsters based on the published FiveThirtyEight methodology. By modifying and

incorporating the measures for historical accuracy and transparency, three models, referred to as rating systems, were established to quantify the reliability of pollsters. Tested against a recent election, all three models demonstrated reasonable accuracy in ranking pollsters, with rating system 1 showing the highest correlation (R_s of 0.893). Moreover, they all predicted the first 3 and the last places correctly, which has significant implications in the aim of the investigation: providing a guide to what pollsters to look at and what not for accurate representations of public sentiment.

However, as with any mathematical model, the pollster rating models are a simplification of reality**.** No model, even the 538's, can fully capture the complexities of political polling due to unquantifiable uncertainties and variabilities from factors like political context, shocks or changes in polling methodologies.

The way in which these models were tested are also imperfect, giving an inherent advantage rating system 1 due to its emphasis in empirical accuracy. As stated in section I, historical accuracy does not guarantee future predictive power, which is the reason transparency and methodological accuracy is considered to account for stability and theoretical soundness. Therefore, this approach used in this exploration leaves room for consideration and improvement on how to account for these factors even in the testing phase.

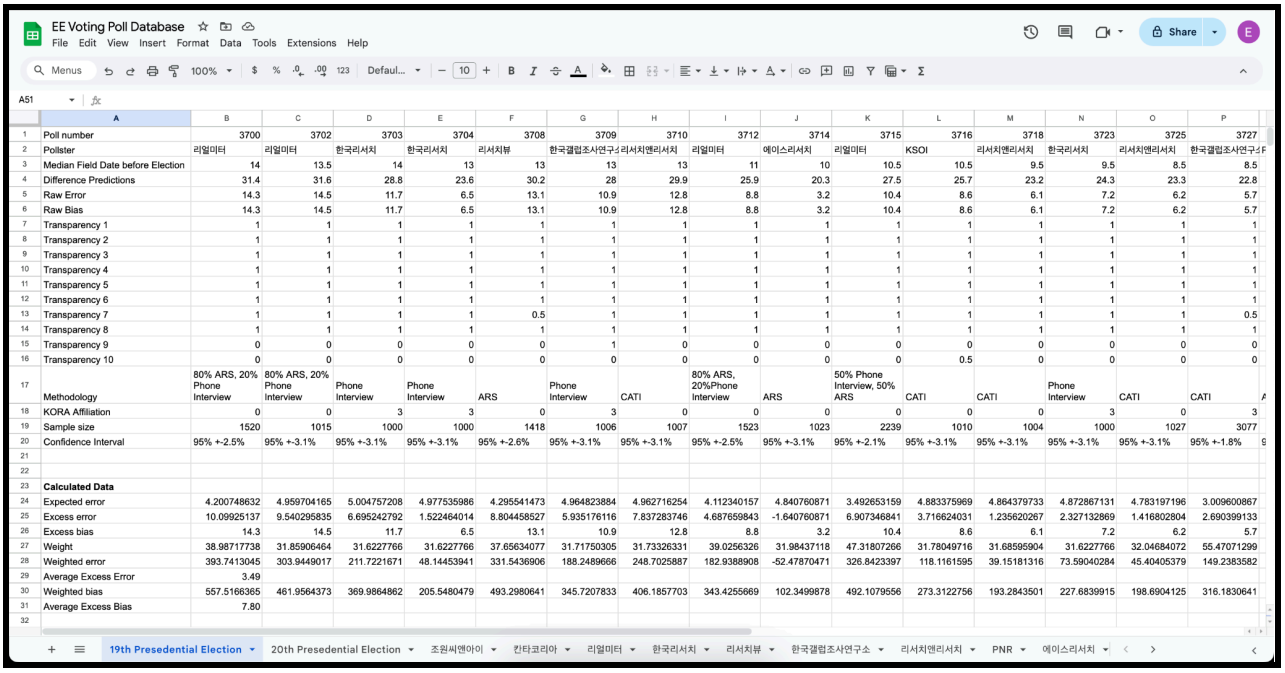
There were also several limitations of the adaptation of the 538 method to the fullest extent, particularly in the calculations process. As noted in section II, guesswork had to be made for many parts, for example the derivation of the expected error formula. While most relations between variables were represented, their degrees were likely inaccurate in many cases. Additionally, the small, limited dataset of 71 polls from two presidential elections restricts the

model's overall reliability and generalizability. Also, certain aspects of FiveThirtyEight's methodology, such as predictive error adjustments and herding effects, were omitted due to data constraints. Future research could refine this model by incorporating machine learning techniques, for 'multilevel regression models' mentioned by the 538, or by analyzing a broader range of elections to improve predictive power.

Despite these limitations, this study provides a grounded mathematical approach to evaluating South Korean pollsters in a data-driven, empirical and replicable manner. While no rating system can fully encapsulate a pollster's quality, this structured model can still offer valuable insights for both policymakers and the public when interpreting polling data.

Word count(excluding tables, titles and equations):

Appendix of Databases created and used

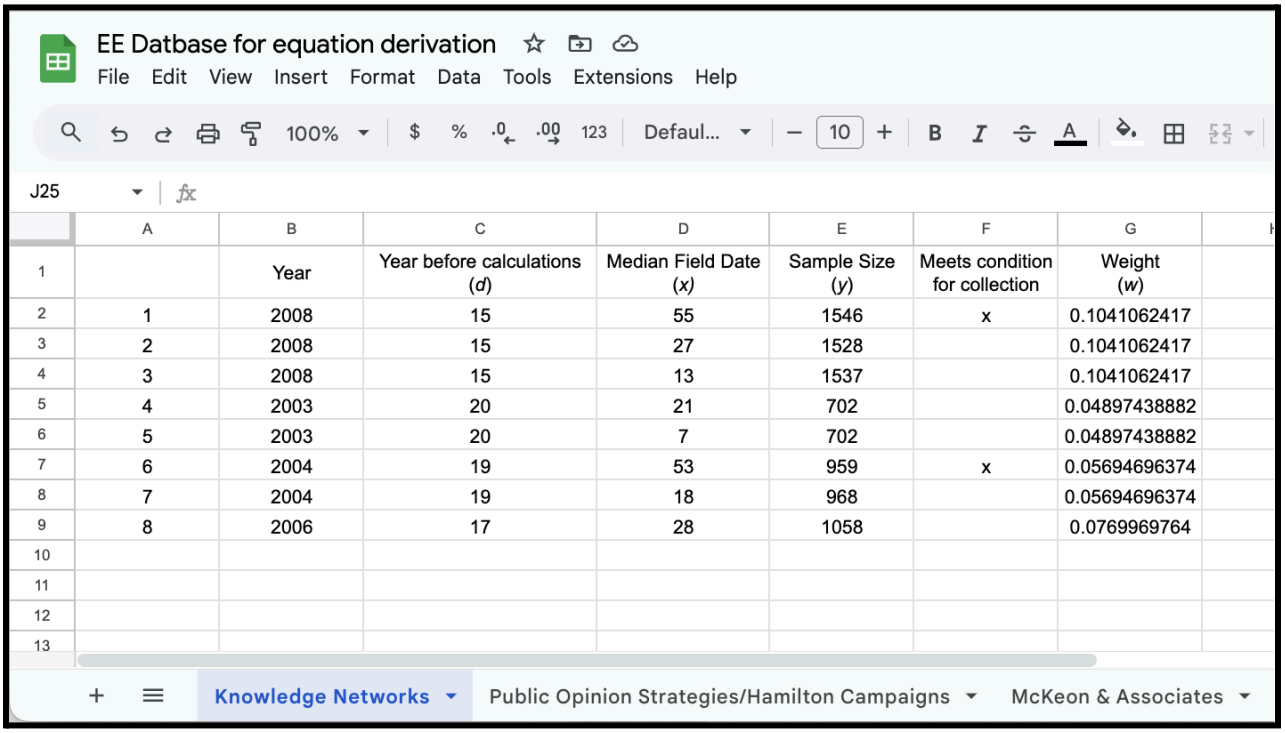


EE Voting Poll Database

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	Poll number	3700	3702	3703	3704	3708	3709	3710	3712	3714	3715	3716	3718	3723	3725	3727
2	Pollster	리얼미터	리얼미터	한국리서치	한국리서치	리서치뷰	한국갤럽조사연구소	리서치앤리서치	리얼미터	에이스리서치	리얼미터	KSOI	리서치앤리서치	한국리서치	리서치앤리서치	한국갤럽조사연구소
3	Median Field Date before Election	14	13.5	14	13	13	13	13	11	10	10.5	10.5	9.5	9.5	8.5	8.5
4	Difference Predictions	31.4	31.6	28.8	23.6	30.2	28	29.9	25.9	20.3	27.5	25.7	23.2	24.3	23.3	22.8
5	Raw Error	14.3	14.5	11.7	6.5	13.1	10.9	12.8	8.8	3.2	10.4	8.6	6.1	7.2	6.2	5.7
6	Raw Bias	14.3	14.5	11.7	6.5	13.1	10.9	12.8	8.8	3.2	10.4	8.6	6.1	7.2	6.2	5.7
7	Transparency 1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
8	Transparency 2	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
9	Transparency 3	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
10	Transparency 4	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
11	Transparency 5	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
12	Transparency 6	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
13	Transparency 7	1	1	1	1	0.5	1	1	1	1	1	1	1	1	1	0.5
14	Transparency 8	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
15	Transparency 9	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0
16	Transparency 10	0	0	0	0	0	0	0	0	0	0	0.5	0	0	0	0
17	Methodology	80% ARS, 20% Phone Interview	80% ARS, 20% Phone Interview	Phone Interview	Phone Interview	ARS	Phone Interview	CATI	80% ARS, 20%Phone Interview	ARS	50% Phone Interview, 50% ARS	CATI	CATI	Phone Interview	CATI	CATI
18	KORA Affiliation	0	0	3	3	0	3	0	0	0	0	0	0	3	0	3
19	Sample size	1520	1015	1000	1000	1418	1006	1007	1523	1023	2239	1010	1004	1000	1027	3077
20	Confidence Interval	95% +-2.5%	95% +-3.1%	95% +-3.1%	95% +-3.1%	95% +-2.6%	95% +-3.1%	95% +-3.1%	95% +-2.5%	95% +-3.1%	95% +-2.1%	95% +-3.1%	95% +-3.1%	95% +-3.1%	95% +-3.1%	95% +-1.8%
21																
22																
23	**Calculated Data**															
24	Expected error	4.200748632	4.959704165	5.004757208	4.977535986	4.295541473	4.964823884	4.962716254	4.112340157	4.840760871	3.492653159	4.883375969	4.864379733	4.872867131	4.783197196	3.009600867
25	Excess error	10.09925137	9.540295835	6.695242792	1.522464014	8.804458527	5.935176116	7.837283746	4.687659843	-1.640760871	6.907346841	3.716624031	1.235620267	2.327132869	1.416802804	2.690399133
26	Excess bias	14.3	14.5	11.7	6.5	13.1	10.9	12.8	8.8	3.2	10.4	8.6	6.1	7.2	6.2	5.7
27	Weight	38.98717738	31.85906464	31.6227766	31.6227766	37.65634077	31.71750305	31.73326331	39.0256326	31.98437118	47.31807266	31.78049716	31.68595904	31.6227766	32.04684072	55.47071299
28	Weighted error	393.7413045	303.9449017	211.7221671	48.14453941	331.5436906	188.2489666	248.7025887	182.9388908	-52.47870471	326.8423397	118.1161595	39.15181316	73.59040284	45.40405379	149.2383582
29	Average Excess Error	3.49														
30	Weighted bias	557.5166365	461.9564373	369.9864862	205.5480479	493.2980641	345.7207833	406.1857703	343.4255669	102.3499878	492.1079556	273.3122756	193.2843501	227.6839915	198.6904125	316.1830641
31	Average Excess Bias	7.80														
32																

19th Presedential Election | 20th Presedential Election | 조원씨앤아이 | 칸타코리아 | 리얼미터 | 한국리서치 | 리서치뷰 | 한국갤럽조사연구소 | 리서치앤리서치 | PNR | 에이스리서치

Excerpt of all polls database - sorted by elections, then pollsters. Pollster names written in Korean



EE Datbase for equation derivation

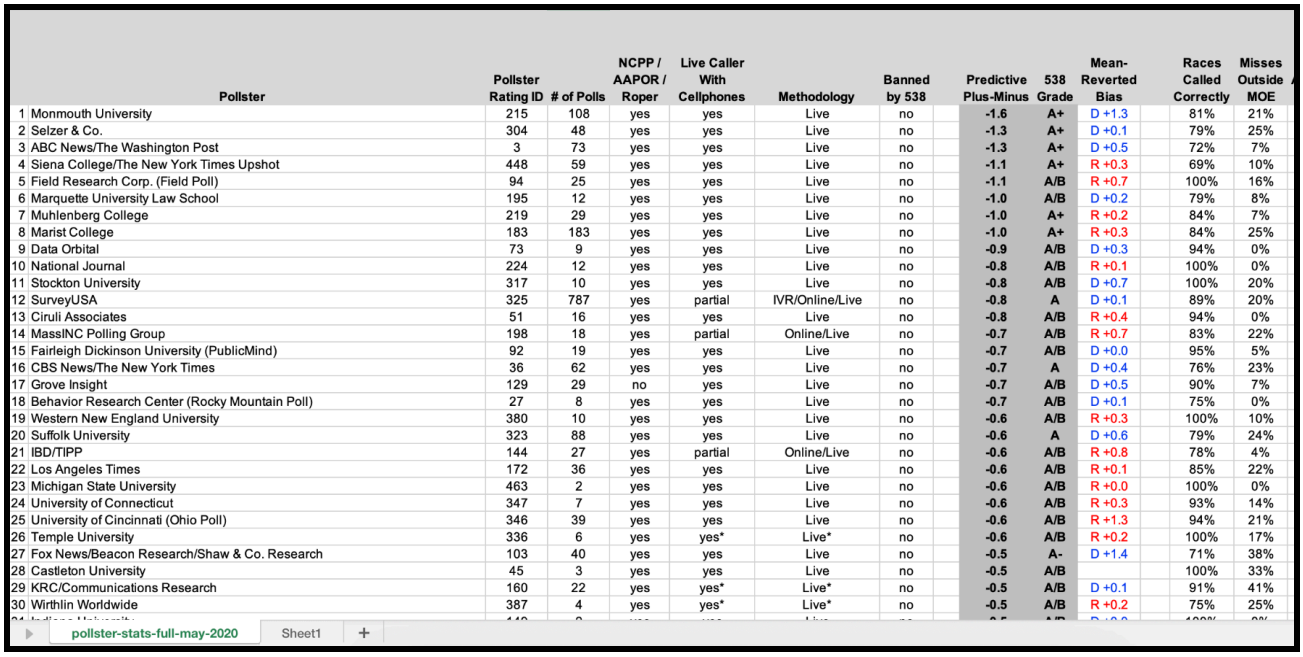
File Edit View Insert Format Data Tools Extensions Help

J25

	A	B	C	D	E	F	G
1		Year	Year before calculations (*d*)	Median Field Date (*x*)	Sample Size (*y*)	Meets condition for collection	Weight (*w*)
2	1	2008	15	55	1546	x	0.1041062417
3	2	2008	15	27	1528		0.1041062417
4	3	2008	15	13	1537		0.1041062417
5	4	2003	20	21	702		0.04897438882
6	5	2003	20	7	702		0.04897438882
7	6	2004	19	53	959	x	0.05694696374
8	7	2004	19	18	968		0.05694696374
9	8	2006	17	28	1058		0.0769969764
10							
11							
12							
13							

Knowledge Networks | Public Opinion Strategies/Hamilton Campaigns | McKeon & Associates

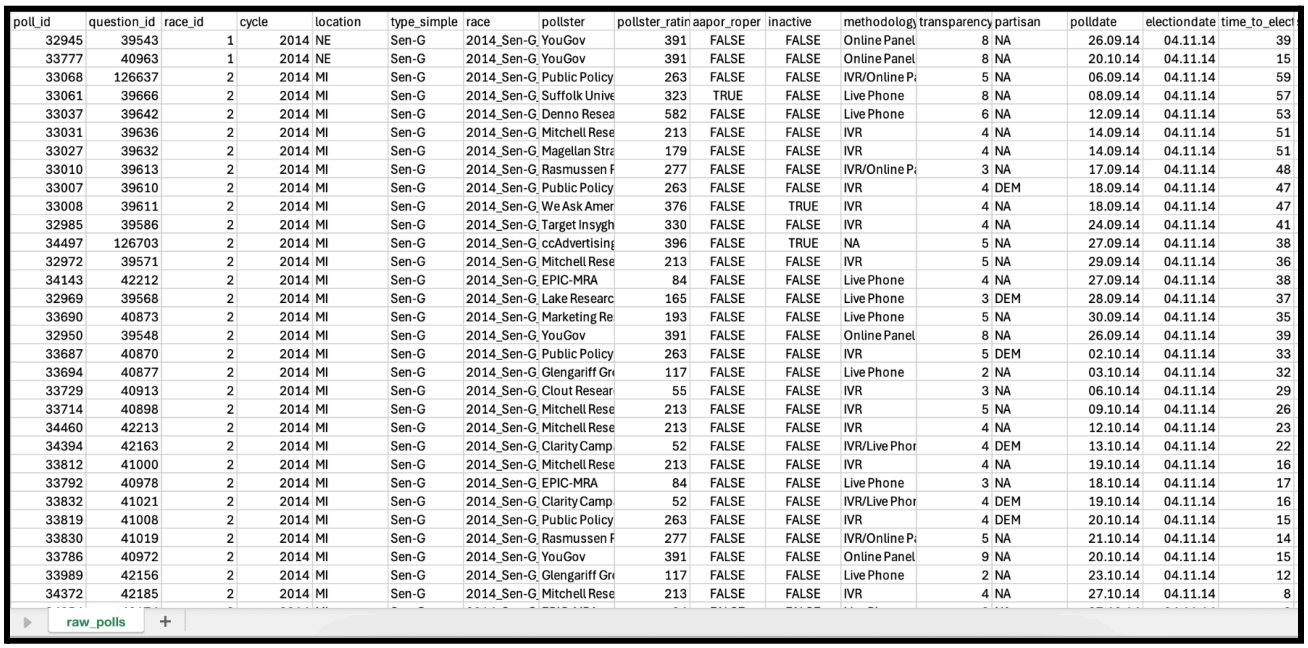
Excerpt of database of sorted polls of 3 pollsters for Expected Error formula derivation, from 538



	Pollster	Pollster Rating ID	# of Polls	NCPP / AAPOR / Roper	Live Caller With Cellphones	Methodology	Banned by 538	Predictive Plus-Minus	538 Grade	Mean-Reverted Bias	Races Called Correctly	Misses Outside MOE
1	Monmouth University	215	108	yes	yes	Live	no	-1.6	A+	D +1.3	81%	21%
2	Selzer & Co.	304	48	yes	yes	Live	no	-1.3	A+	D +0.1	79%	25%
3	ABC News/The Washington Post	3	73	yes	yes	Live	no	-1.3	A+	D +0.5	72%	7%
4	Siena College/The New York Times Upshot	448	59	yes	yes	Live	no	-1.1	A+	R +0.3	69%	10%
5	Field Research Corp. (Field Poll)	94	25	yes	yes	Live	no	-1.1	A/B	R +0.7	100%	16%
6	Marquette University Law School	195	12	yes	yes	Live	no	-1.0	A/B	D +0.2	79%	8%
7	Muhlenberg College	219	29	yes	yes	Live	no	-1.0	A+	R +0.2	84%	7%
8	Marist College	183	183	yes	yes	Live	no	-1.0	A+	R +0.3	84%	25%
9	Data Orbital	73	9	yes	yes	Live	no	-0.9	A/B	D +0.3	94%	0%
10	National Journal	224	12	yes	yes	Live	no	-0.8	A/B	R +0.1	100%	0%
11	Stockton University	317	10	yes	yes	Live	no	-0.8	A/B	D +0.7	100%	20%
12	SurveyUSA	325	787	yes	partial	IVR/Online/Live	no	-0.8	A	D +0.1	89%	20%
13	Ciruli Associates	51	16	yes	yes	Live	no	-0.8	A/B	R +0.4	94%	0%
14	MassINC Polling Group	198	18	yes	partial	Online/Live	no	-0.7	A/B	R +0.7	83%	22%
15	Fairleigh Dickinson University (PublicMind)	92	19	yes	yes	Live	no	-0.7	A/B	D +0.0	95%	5%
16	CBS News/The New York Times	36	62	yes	yes	Live	no	-0.7	A	D +0.4	76%	23%
17	Grove Insight	129	29	no	yes	Live	no	-0.7	A/B	D +0.5	90%	7%
18	Behavior Research Center (Rocky Mountain Poll)	27	8	yes	yes	Live	no	-0.7	A/B	D +0.1	75%	0%
19	Western New England University	380	10	yes	yes	Live	no	-0.6	A/B	R +0.3	100%	10%
20	Suffolk University	323	88	yes	yes	Live	no	-0.6	A	D +0.6	79%	24%
21	IBD/TIPP	144	27	yes	partial	Online/Live	no	-0.6	A/B	R +0.8	78%	4%
22	Los Angeles Times	172	36	yes	yes	Live	no	-0.6	A/B	R +0.1	85%	22%
23	Michigan State University	463	2	yes	yes	Live	no	-0.6	A/B	R +0.0	100%	0%
24	University of Connecticut	347	7	yes	yes	Live	no	-0.6	A/B	R +0.3	93%	14%
25	University of Cincinnati (Ohio Poll)	346	39	yes	yes	Live	no	-0.6	A/B	R +1.3	94%	21%
26	Temple University	336	6	yes	yes*	Live*	no	-0.6	A/B	R +0.2	100%	17%
27	Fox News/Beacon Research/Shaw & Co. Research	103	40	yes	yes	Live	no	-0.5	A-	D +1.4	71%	38%
28	Castleton University	45	3	yes	yes	Live	no	-0.5	A/B		100%	33%
29	KRC/Communications Research	160	22	yes	yes*	Live*	no	-0.5	A/B	D +0.1	91%	41%
30	Wirthlin Worldwide	387	4	yes	yes*	Live*	no	-0.5	A/B	R +0.2	75%	25%

pollster-stats-full-may-2020 | Sheet1 | +

Excerpt of database from 538 of pollster statistics and data (file name: pollster-stats-full)



poll_id	question_id	race_id	cycle	location	type_simple	race	pollster	pollster_ratin	aapor_roper	inactive	methodology	transparency	partisan	polldate	electiondate	time_to_elect
32945	39543	1	2014	NE	Sen-G	2014_Sen-G	YouGov	391	FALSE	FALSE	Online Panel	8	NA	26.09.14	04.11.14	39
33777	40963	1	2014	NE	Sen-G	2014_Sen-G	YouGov	391	FALSE	FALSE	Online Panel	8	NA	20.10.14	04.11.14	15
33068	126637	2	2014	MI	Sen-G	2014_Sen-G	Public Policy	263	FALSE	FALSE	IVR/Online Pa	5	NA	06.09.14	04.11.14	59
33061	39666	2	2014	MI	Sen-G	2014_Sen-G	Suffolk Unive	323	TRUE	FALSE	Live Phone	8	NA	08.09.14	04.11.14	57
33037	39642	2	2014	MI	Sen-G	2014_Sen-G	Denno Resea	582	FALSE	FALSE	Live Phone	6	NA	12.09.14	04.11.14	53
33031	39636	2	2014	MI	Sen-G	2014_Sen-G	Mitchell Rese	213	FALSE	FALSE	IVR	4	NA	14.09.14	04.11.14	51
33027	39632	2	2014	MI	Sen-G	2014_Sen-G	Magellan Stra	179	FALSE	FALSE	IVR	4	NA	14.09.14	04.11.14	51
33010	39613	2	2014	MI	Sen-G	2014_Sen-G	Rasmussen F	277	FALSE	FALSE	IVR/Online Pa	3	NA	17.09.14	04.11.14	48
33007	39610	2	2014	MI	Sen-G	2014_Sen-G	Public Policy	263	FALSE	FALSE	IVR	4	DEM	18.09.14	04.11.14	47
33008	39611	2	2014	MI	Sen-G	2014_Sen-G	We Ask Amer	376	FALSE	TRUE	IVR	4	NA	18.09.14	04.11.14	47
32985	39586	2	2014	MI	Sen-G	2014_Sen-G	Target Insygh	330	FALSE	FALSE	IVR	4	NA	24.09.14	04.11.14	41
34497	126703	2	2014	MI	Sen-G	2014_Sen-G	ccAdvertising	396	FALSE	TRUE	NA	5	NA	27.09.14	04.11.14	38
32972	39571	2	2014	MI	Sen-G	2014_Sen-G	Mitchell Rese	213	FALSE	FALSE	IVR	5	NA	29.09.14	04.11.14	36
34143	42212	2	2014	MI	Sen-G	2014_Sen-G	EPIC-MRA	84	FALSE	FALSE	Live Phone	4	NA	27.09.14	04.11.14	38
32969	39568	2	2014	MI	Sen-G	2014_Sen-G	Lake Researc	165	FALSE	FALSE	Live Phone	3	DEM	28.09.14	04.11.14	37
33690	40873	2	2014	MI	Sen-G	2014_Sen-G	Marketing Re	193	FALSE	FALSE	Live Phone	5	NA	30.09.14	04.11.14	35
32950	39548	2	2014	MI	Sen-G	2014_Sen-G	YouGov	391	FALSE	FALSE	Online Panel	8	NA	26.09.14	04.11.14	39
33687	40870	2	2014	MI	Sen-G	2014_Sen-G	Public Policy	263	FALSE	FALSE	IVR	5	DEM	02.10.14	04.11.14	33
33694	40877	2	2014	MI	Sen-G	2014_Sen-G	Glengariff Gr	117	FALSE	FALSE	Live Phone	2	NA	03.10.14	04.11.14	32
33729	40913	2	2014	MI	Sen-G	2014_Sen-G	Clout Resear	55	FALSE	FALSE	IVR	3	NA	06.10.14	04.11.14	29
33714	40898	2	2014	MI	Sen-G	2014_Sen-G	Mitchell Rese	213	FALSE	FALSE	IVR	5	NA	09.10.14	04.11.14	26
34460	42213	2	2014	MI	Sen-G	2014_Sen-G	Mitchell Rese	213	FALSE	FALSE	IVR	4	NA	12.10.14	04.11.14	23
34394	42163	2	2014	MI	Sen-G	2014_Sen-G	Clarity Camp	52	FALSE	FALSE	IVR/Live Phor	4	DEM	13.10.14	04.11.14	22
33812	41000	2	2014	MI	Sen-G	2014_Sen-G	Mitchell Rese	213	FALSE	FALSE	IVR	4	NA	19.10.14	04.11.14	16
33792	40978	2	2014	MI	Sen-G	2014_Sen-G	EPIC-MRA	84	FALSE	FALSE	Live Phone	3	NA	18.10.14	04.11.14	17
33832	41021	2	2014	MI	Sen-G	2014_Sen-G	Clarity Camp	52	FALSE	FALSE	IVR/Live Phor	4	DEM	19.10.14	04.11.14	16
33819	41008	2	2014	MI	Sen-G	2014_Sen-G	Public Policy	263	FALSE	FALSE	IVR	4	DEM	20.10.14	04.11.14	15
33830	41019	2	2014	MI	Sen-G	2014_Sen-G	Rasmussen F	277	FALSE	FALSE	IVR/Online Pa	5	NA	21.10.14	04.11.14	14
33786	40972	2	2014	MI	Sen-G	2014_Sen-G	YouGov	391	FALSE	FALSE	Online Panel	9	NA	20.10.14	04.11.14	15
33989	42156	2	2014	MI	Sen-G	2014_Sen-G	Glengariff Gr	117	FALSE	FALSE	Live Phone	2	NA	23.10.14	04.11.14	12
34372	42185	2	2014	MI	Sen-G	2014_Sen-G	Mitchell Rese	213	FALSE	FALSE	IVR	4	NA	27.10.14	04.11.14	8

raw_polls | +

Excerpt of database from 538 of raw poll statistics and data (file name: raw_polls)

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